

BUDHADITYA MUKHOPADHYAY

Senior AI Engineer · GenAI & Edge AI · Strategic Advisor — UAV & AI Innovation

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≈97%	~16 MB	~200 ms	2	3	10	9+
MODEL ACCURACY	ON-DEVICE MODEL	INFERENCE TIME	CLINICAL TRIALS	PUBLICATIONS	MENTEES	YEARS EXP.

SUMMARY

Senior AI Engineer in Munich specializing in production AI systems: healthcare ML, GenAI, edge/on-device inference, audio & video AI, computer vision, and AI-powered UAV systems. End-to-end ownership from data strategy and model development to validation, deployment, and mobile integration — backed by 3 peer-reviewed publications in medical AI and explainability, and hands-on mentoring of 10 students.

SKILLS

AI / ML	PyTorch, TensorFlow, TensorFlow.js, Computer Vision, Audio AI, Medical Imaging, Temporal Modeling, Explainable AI (XAI)
GenAI / LLM	Ollama, OpenAI-compatible APIs, LLM integration, agent workflows, FastAPI, Gradio
Edge / Mobile	On-device & real-time inference, model optimization, React Native
MLOps / Backend	Docker, CI/CD, multi-GPU training, Flask, data pipelines, API development
Languages	Python, TypeScript, JavaScript, C++ (exposure)

EXPERIENCE

Senior AI Engineer Feb 2026 – Present

[VisionHealth GmbH · Munich, Germany](#) (AI Engineer: Jan 2023 – Jan 2026)

- Own the full AI architecture for audio/video healthcare AI as the team's sole AI engineer — data strategy, model training, validation, and production deployment.
- Built an inhalation-detection model with ≈97% accuracy, deployed on-device at ~16 MB with ~200 ms real-time inference (YAMNet audio feature extraction).
- Shipped TensorFlow.js models detecting inhalation, spray, exhalation, and manoeuvre quality into React Native mobile pipelines.
- Supported two clinical trials (pharma company Sterna; the Kata app with ~2,300 users) with framework-based and human-in-the-loop validation.
- Collected and curated diverse training data (2,000+ inhaler images, ~3,000 audio files across devices, OS versions, and demographics).
- Mentored 3 master students and taught AI / AI ethics to colleagues.

Strategic Advisor May 2025 – Present

[Weevils Drones](#)

- Designed the end-to-end software architecture for AI-powered drone and anti-drone systems, bridging hardware, software, and ML into operational platforms.
- Advise on AI model selection across detection, tracking, classification, localization, threat scoring, path planning, and autonomy; sensor stack spans IMU, GPS, magnetometer, barometer, LiDAR, and camera (Python / C++ / C#).
- Develop drone-to-GUI connectivity software; mentor 7 interns and advise leadership on strategy, partnerships, and go-to-market.

Data Science Researcher (Master Thesis) & Intern Jan 2022 – Dec 2022

[Volkswagen AG · Wolfsburg, Germany](#)

- Developed deep-learning motion-sickness detection in vehicles from non-contact features — pupil-diameter time series (LSTM, 1D CNN), facial data, and contextual data with explainability.
- Identified facial data, pupil diameter, and MSSQ-short as promising predictive signals for in-car motion sickness.

Student Research Assistant

Nov 2020 – Sep 2022

Otto-von-Guericke University Magdeburg, Germany

- Built a deep-learning application for blur detection in MRI/CT scans with 2D/3D pipelines, multi-GPU and multi-threaded training, and XAI-based interpretation.
- Developed a PyTorch dataset-distribution tool (text/audio/image) and a German speech-to-text translation service (Flask, Docker, CI/CD).

Research Assistant

Aug 2020 – Oct 2021

DZNE — German Center for Neurodegenerative Diseases

- Created an end-to-end video pipeline for ALS bulbar-function quantification: denoising, face/lip tracking (FaceNet), tongue segmentation (MobileNetV2, YOLO, RCNN), active-learning dataset growth, Flask GUI, and Docker packaging.
- Work contributed to a Frontiers in Neurology publication (co-first author).

Associate — Pharmacy-domain APIs & validation automation

Oct 2018 – Sep 2019

Cognizant · Kolkata, India

Solution Integrator — Telecom integration; onsite SPOC (Kuwait)

Feb 2015 – Oct 2018

Ericsson · India / Kuwait

EDUCATION

M.Sc. Data and Knowledge Engineering

2019 – 2023

Otto-von-Guericke University Magdeburg, Germany — Thesis with Volkswagen AG (motion sickness detection)

B.Tech. Information Technology

2010 – 2014

West Bengal University of Technology, Kolkata, India

PUBLICATIONS

First author Towards Segmenting the Invisible: An End-to-End Registration and Segmentation Framework for Weakly Supervised Tumour Analysis

AIBIO 2025 · Springer CCIS vol. 2696, pp. 229–242 · doi:10.1007/978-3-032-17216-7_18

Co-first author An Automated Tongue Tracker for Quantifying Bulbar Function in ALS

Frontiers in Neurology, vol. 13, 2022 · doi:10.3389/fneur.2022.838191

Co-author TorchEsegeta: Framework for Interpretability and Explainability of Image-Based Deep Learning Models

Applied Sciences, 12(4), 1834, 2022 · doi:10.3390/app12041834

SELECTED GENAI PROJECTS

RemoteLlama — local-first Next.js chat UI for remote Ollama with streaming and document ingestion. [GitHub](#)

AI Repo Analyzer — FastAPI + Gradio GitHub-repository analysis with LLM enrichment and report export. [GitHub](#)

CareerCrawler — AI job-intelligence platform with skill-gap analysis and guarded automation. [GitHub](#)

JarvisFlow — gesture-controlled brainstorming app using in-browser MediaPipe hand tracking. [GitHub](#)

LANGUAGES & TEACHING

English (native) · Bengali (native) · Hindi (fluent) · German (preparing B1). Main speaker, "Agents in Action: The Dawn of Action-Oriented AI" webinar (~60 students); mentor to 10 students across VisionHealth and Weevils Drones.